Indoor Mobile Robot Localization and Mapping Based on Ambient Magnetic Fields and Aiding Radio Sources

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***Abstract*— In robotics, the problem of concurrently addressing the localization and mapping is well defined as simultaneous localization and mapping (SLAM) problem. Since the SLAM procedure is usually recursive, maintaining a certain error bound on the current position estimate is a critical issue. However, when the robot is kidnapped (i.e., the robot is moved by an intentional or unintentional user) or suffers from locomotion failure (due to large slip and falling), the robot will inevitably lose its current position. In this case, immediate recovery of the robot position is essential for seamless operation. In this paper, we present a method of solving both SLAM and relocation problems by employing ambient magnetic and radio measurements. The proposed SLAM is realized in the Rao-Blackwellized particle filter- and grid-based SLAM frameworks, where we exploit the local heading corrections from the magnetic measurements. For the relocation, we design the location signatures using the magnetic and radio measurements, and examine each of the Monte Carlo localization-based and multilayer perceptron-based relocation methods with real-world data. We implement the proposed SLAM and relocation algorithms in an embedded system and verify the feasibility of the proposed methods as an online robot navigation system.**

***Index Terms*— Geomagnetic field, radio ranging, relocation, simultaneous localization and mapping (SLAM).**

I. Introduction

MOBILE robot localization is the problem of estimating the pose of a robot with respect to a given map of the environment. Without the aid of GPS, which is substantial, localization in indoor environments is far more sophisticated than the outdoors. In general, a robot in an indoor navigation task is capable of sensing its movement and the surrounding environment. Based on the gathered sensory observations, the robot can obtain its current position by maximizing the likelihood of that observations given prior knowledge of the map [1], [2].

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In real-world applications, however, obtaining a precise indoor map in advance is often not practical and sometimes impossible (e.g., in a task of exploration). Therefore, being able to build a map from scratch is another competency of the truly autonomous robots. In robotics, the problem of concurrently addressing the localization and mapping is well defined as a simultaneous localization and mapping (SLAM) problem. The SLAM problem has been intensively researched and addressed by many novel and high-performing methodologies [1], [3]–[5].

Since the SLAM procedure is usually recursive, maintaining a certain error bound on the current position estimate is a critical issue. However, when the robot is kidnapped (i.e., the robot is moved by an intentional or unintentional user) or suffers from locomotion failure (due to large slip and falling), the robot will inevitably lose its current position [6], [7]. In this case, immediate recovery of the robot position is essential for seamless operation. Several works have been carried out to solve related problems either by fault diagnosis using adaptive filtering methods [8], [9] or structural adaptation using multiple models [6], [10]–[12]. However, the adaptation approaches cannot guarantee error convergence in severe conditions and in that case, reinitialization of the robot state is required. This motivates our work to focus on solving both SLAM and global relocation problem.

In general, the SLAM framework is designed to allow a variety of external sensors, such as cameras, laser range finders, and radio frequency beacons. Key process in SLAM is then to extract the location-related features from the sensory data and manage these as pertinent map components called landmarks. In this paper, we employ the Rao-Blackwellized particle filter (RBPF) [13] and grid-based landmark construction [14] as our basic SLAM framework. In this framework, we are trying to build landmarks from ambient signal sources, especially from geomagnetic fields and radio ranging signals.

In classic inertial or range-based navigation systems, geomagnetic fields and radio ranges have not been much of use for the indoor environments. This is because the geomagnetic field is easily distorted by ferromagnetic objects [15] and the radio signal suffers from the attenuation and multipath effect [16]. Therefore, many works have tried to exclude or mitigate the abnormal observations of these signals [17]–[21]. In this paper, however, we consider the magnetic anomalies and the noisy radio ranges as unique signatures of the robot's location. Specific to the magnetic measurement, many works have proved the feasibility of using it as indoor location

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signatures [22]–[27]. Later in this paper, it will be shown that using the magnetic source alone is sufficient for the SLAM, while in the relocation phase, additional radio sources are helpful to resolve the ambiguity issues of the magnetic sources.

For the global relocation framework, we tried two representative approaches from probabilistic inference and machine learning techniques. We used Monte Carlo localization (MCL) and multilayer perceptron (MLP) to realize our relocation solution and evaluated each one's performance. We expect that if the position estimation is sufficiently accurate, the orientation can be converged by moving the robot along a certain distance. Therefore, the proposed relocation algorithms aim to recover the robot's 2-D position only.

As for the contribution of this paper, there is no existing method that solves a combined problem of SLAM and global relocation using magnetic measurements. Applying noisy radio range measurements into the location-signature-based localization is also a new one. The approaches of RBPF, MCL, and MLP used in our work are not completely new, but we modified the update equations of RBPF and MCL and designed novel input features for MLP to consider pertinent measurement characteristics.

The rest of this paper is organized as follows. Section II contains the related work. Sections III and IV describe the proposed magnetic SLAM and relocation algorithms, respectively. Section V validates the proposed algorithms with simulation and experimental results. Finally, the conclusion is drawn in Section VI.

II. Related Work

Approaches that apply magnetic fields to the positioning problem can be categorized into two groups according to whether they are based on the active magnetic fields or passive ambient magnetic fields. In the former, magnetic fields are artificially generated in the area of interest using active beacons. Empirical magnetic field models then can be used to predict the observed magnetic strength and the relevant distance values [28]–[30]. Especially, Sheinker *et al.* [30] installed beacons generating low-frequency magnetic field and derived receiver location in a closed form using a theoretical magnetic field model. They have achieved mean positioning error of less than 0.25 m in a laboratory environment. These active beacon-based approaches have the advantage that the magnetic field signals are strong and controllable but require active magnetic sources, such as coil infrastructures or permanent magnets. Moreover, empirical models cannot consider the magnetic distortions caused by the environments. The latter, ambient magnetic field-based positioning, is growing in popularity due to the wide diffusion of mobile phones equipped with electronic compasses. An electronic compass measures three-axis geomagnetic field values to obtain the global heading information. In indoor environments, however, the magnetic fields are usually distorted by other electronic devices or ferromagnetic objects, and this adversely affects the performance of the compass. To overcome this problem, adaptive filtering techniques such as those in [31] can be used. In [31], magnetic disturbance has been mitigated by adapting the Kalman filter's measurement covariance using the measured strength of the magnetic field vector.

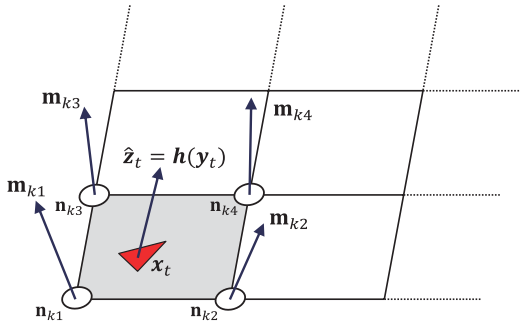


Fig. 1. State representation of robot and map. Shaded area represents the current cell where the robot is located.

The presence of anomalies in the geomagnetic field, in turn, can provide the opportunity of using themselves as location fingerprints. In [22], the magnetic measurements from an array of magnetometers are employed to construct the underground magnetic fingerprints and are applied to the human positioning system. For indoor environments, Gozick *et al.* [15], Li *et al.* [23], Krisp [24], and Chung *et al.* [32] have investigated the magnetic field behavior with an extensive measurement data set gathered with different mobile devices with different sensitivity rates. They also showed the feasibility of applying magnetic fingerprints to the distinction of floor- or room-level indoor locations. The reason for the lower resolution of these systems is that the magnetic signature has a lot of ambiguities. Therefore, in our work, we added additional radio sources to solve the ambiguity problems.

Extensive work has also been done to develop radio beacon-based indoor positioning systems. Among various radio communication signals, ultrawideband [33] and chirp spread spectrum (CSS) [34] signals are especially dedicated to the precise ranging between the transceiver(s) and receiver(s) nodes in a line-of-sight (LOS) condition. Related to these kind of sensor modalities, many works have focused on mitigating the non-LOS (NLOS) ranging errors to enhance the positioning performance. Hirsch *et al.* [35] simulated multiple reflections and diffraction of ranging signals using ray-tracing method and applied this to the LOS/NLOS distinction. Similarly, He *et al.* [16] simulated multipath effects by considering multiple multipath fading channels and evaluated the actual ranging errors. Even these sophisticated simulators, however, cannot consider all the propagation parameters and these results in a large ranging error up to 15 m. In our work, the range measurements (regardless of whether they are LOS or NLOS ranges) are considered as location signatures and combined with magnetic fingerprints to enhance the uniqueness of the fingerprints. In this way, we can exploit the ranges even with large NLOS errors.

III. Magnetic SLAM

A. State Representation

We first define the state variables related to the localization and mapping problems. Fig. 1 shows the robot and map state variables represented in a gridded exploration area. It is

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assumed that a time series of the robot state $\{\mathbf{x}_t\}$ is defined in an $\mathbf{SE}(2)$ pose set as

$$\mathbf{x}_t = [x_t, y_t, \theta_t]^T, \quad t = 0, \ldots, t_f \tag{1}$$

with an initial pose $\mathbf{x}_0 = [0, 0, 0]^T$. In the proposed magnetic SLAM formulation, the planar space of the robot's movement is divided into a finite number of rectangular cells. The map state $\{\mathbf{m}_l\}$ is then defined as a vector describing the geomagnetic field at the fixed node position $\mathbf{n}_l = [n_{l,x}, n_{l,y}]^T$ as follows:

$$\mathbf{m}_l = [m_x, m_y, m_z]^T, \quad l = 1, \ldots, M \tag{2}$$

where $[m_x, m_y\ m_z]^T$ is a magnetic triad represented in a local reference frame defined at the robot's initial pose and M is the total number of the explored nodes. We assume that no significant changes in the magnetic field occur during the robot's temporal operation. Therefore, the magnetic landmarks are treated as static states. Similarly, the measurement $\mathbf{z}_t$ can be defined as a magnetic triad represented in a sensor-fixed reference frame as follows:

$$\mathbf{z}_t = [B_x, B_y, B_z]^T, \quad t = 0, \ldots, t_f \tag{3}$$

where B_x, B_y, and B_z are the three orthogonal components of the magnetic field vector. We assume that the robot rotates on a horizontal plane defined by geographic north and east, which makes the vertical component of the measurement vector the same in both local and sensor-fixed reference frames. This assumption makes the z-component of the geomagnetic field independent of robot orientation, which will be exploited in our relocation algorithm. Finally, the augmented state variable $\mathbf{y}_t$ is defined as follows:

$$\mathbf{y}_t = \begin{bmatrix} \mathbf{x}_t \\ \mathbf{m}_1 \\ \vdots \\ \mathbf{m}_M \end{bmatrix}. \tag{4}$$

B. SLAM Formulation

In the RBPF framework, the particles $\{\mathbf{y}_t^i\}$ containing all the state variables and the relevant weight value β^i are generated first. After proper initialization, two key steps for the SLAM procedure—prediction and measurement update are performed in a recursive manner. In the following descriptions, the particle index i is omitted for simplicity of the representation.

1) Prediction: In the prediction step, the state transition is predicted for each particle based on the state transition model $\mathbf{f}(\mathbf{y}_{t-1}, \mathbf{u}_t)$. Usually, the robot kinematics is used to describe this model. Assuming a two-wheeled differential drive robot, the kinematic equations are defined as follows:

$$\begin{aligned} \mathbf{y}_t &= \mathbf{f}(\mathbf{y}_{t-1}, \mathbf{u}_t) \\ &= \begin{bmatrix} x_{t-1} + \frac{1}{2}(u_L + u_R)\Delta t \cos\theta_{t-1} \\ y_{t-1} + \frac{1}{2}(u_L + u_R)\Delta t \sin\theta_{t-1} \\ \theta_{t-1} + \frac{1}{D}(u_R - u_L)\Delta t \\ \mathbf{m}_1 \\ \vdots \\ \mathbf{m}_M \end{bmatrix} \end{aligned} \tag{5}$$

where $\mathbf{u}_t = [u_L, u_R]^T$ are the left and right wheel velocities with a noise covariance $\mathbf{Q}$, D is the distance between two wheels, and Δt is the time interval between the current and previous time step. In (5), only the upper three elements (the robot state) undergo the state transition and thus are updated. The map states remain unchanged in the prediction step.

2) Measurement Update: In the measurement update step, both the robot and the map states are updated according to the measurement model $\mathbf{h}(\mathbf{y}_t)$. As explained in the previous section, the measurement $\mathbf{z}_t$ is represented in a sensor-fixed reference frame and is related to the sensor's orientation. The predicted measurement $\hat{\mathbf{z}}_t$ then can be modeled as follows:

$$\begin{aligned} \hat{\mathbf{z}}_t &= \mathbf{h}(\mathbf{y}_t) \\ &= \mathbf{h}_R\big(\mathbf{h}_0(x_t, y_t, \mathbf{m}_1, \ldots, \mathbf{m}_M), \theta_t\big) \end{aligned} \tag{6}$$

where $\mathbf{h}_0$ and $\mathbf{h}_R$ are the measurement prediction with predefined sensor orientation (i.e., $\theta = 0$) and with actual robot orientation, respectively. First, the $\mathbf{h}_0$ is calculated by the weighted sum of the map states as follows:

$$\mathbf{h}_0 = \sum_{j \in \mathcal{K}} w_j(x_t, y_t, \mathbf{n}_j)\mathbf{m}_j \tag{7}$$

where $w_j(\cdot)$ is a weighting kernel and $\mathcal{K}$ is a set of pertinently chosen map state indices. In this paper, we employed bilinear interpolation as a weighting kernel [14] and the $\mathcal{K}$ becomes a set of indices corresponding to the four nodes surrounding the current robot pose (Fig. 1). We simply set $\mathcal{K} = \{l, l+1, l+2, l+3\}$ in the following formulations, although actual implementation requires more sophisticated handling of these indices. The $\mathbf{h}_R$ is then computed by the 2-D rotation of $\mathbf{h}_0$ as follows:

$$\mathbf{h}_R = \begin{bmatrix} \cos\theta_t & \sin\theta_t & 0 \\ -\sin\theta_t & \cos\theta_t & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{h}_0. \tag{8}$$

In the RBPF framework, the update of the map states is handled by a Kalman filtering. We first calculate the Jacobian $\mathbf{H}_\mathbf{m}$ of the measurement model with respect to the map states

$$\begin{aligned} \mathbf{H}_\mathbf{m} &= \frac{\partial \mathbf{h}}{\partial \mathbf{m}}(\mathbf{y}_t) \\ &= \begin{bmatrix} \mathbf{H}_{\mathbf{m}_l} & \mathbf{H}_{\mathbf{m}_{l+1}} & \mathbf{H}_{\mathbf{m}_{l+2}} & \mathbf{H}_{\mathbf{m}_{l+3}} \end{bmatrix} \end{aligned} \tag{9}$$

$$\mathbf{H}_{\mathbf{m}_j} = \begin{bmatrix} w_j \cos\theta_t & w_j \sin\theta_t & 0 \\ -w_j \sin\theta_t & w_j \cos\theta_t & 0 \\ 0 & 0 & w_j \end{bmatrix} \tag{10}$$

where $\mathbf{H}_{\mathbf{m}_j}$ is a Jacobian block corresponding to the jth map state. With this Jacobian matrix, the Kalman update is performed as follows:

$$\mathbf{S}_t = \mathbf{H}_\mathbf{m}\mathbf{P}_{t-1}\mathbf{H}_\mathbf{m}^T + \mathbf{R} \tag{11}$$

$$\mathbf{K}_t = \mathbf{P}_{t-1}\mathbf{H}_\mathbf{m}^T\mathbf{S}_t^{-1} \tag{12}$$

$$\mathbf{P}_t = \mathbf{P}_{t-1} - \mathbf{K}_t\mathbf{S}_t\mathbf{K}_t^T \tag{13}$$

$$\begin{bmatrix} \mathbf{m}_l \\ \vdots \\ \mathbf{m}_{l+3} \end{bmatrix}_t = \begin{bmatrix} \mathbf{m}_l \\ \vdots \\ \mathbf{m}_{l+3} \end{bmatrix}_{t-1} + \mathbf{K}_t(\mathbf{z}_t - \hat{\mathbf{z}}_t) \tag{14}$$

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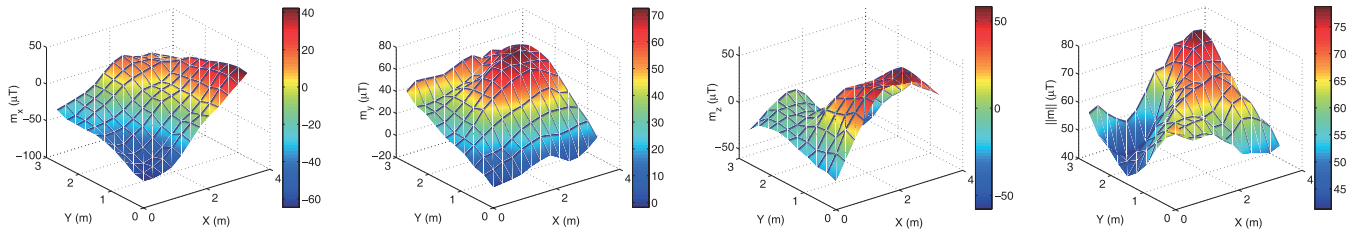


Fig. 2. Indoor geomagnetic fields constructed by the magnetic SLAM. All units are in μT. (a)–(c) Three components of the magnetic field described in a local reference frame. (d) Total strength of the magnetic field. The plot was generated by bilinear interpolation of the estimated map states. It can be observed that the magnetic field is not bijective, which gives rise to location ambiguities between similar fields.

where $\mathbf{P}_t$ is a (12×12) map state error covariance matrix, $\mathbf{R}$ is a (3×3) measurement error covariance matrix, $\mathbf{K}_t$ is a Kalman gain matrix, and $\mathbf{S}_t$ is an innovation covariance matrix.

The robot state is then updated by updating the weight of each particle

$$\beta_t^i = \beta_{t-1}^i \frac{1}{\sqrt{2\pi\left|\mathbf{S}_t^i\right|}} \exp\left[-\frac{1}{2}\boldsymbol{\nu}_t^{iT}\left(\mathbf{S}_t^i\right)^{-1}\boldsymbol{\nu}_t^i\right] \tag{15}$$

where $\boldsymbol{\nu}_t^i = \mathbf{z}_t^i - \hat{\mathbf{z}}_t^i$ is the measurement residual for the ith particle. Resampling is performed whenever the size of effective particles becomes lower than a certain threshold [1]. With the updated particle set, the final estimated state at time step t is calculated as follows:

$$\hat{\mathbf{y}}_t = \sum_{i=1}^{N_p} \mathbf{y}_t^i \beta_t^i \tag{16}$$

where N_p is the number of particles used in the RBPF.

IV. Relocation

A. Design of Fingerprints

For successful relocation, a location signature called fingerprint should be carefully designed in advance by examining the characteristics of the available signal sources. The designed fingerprints then will be encoded during the SLAM procedure and will be used whenever a global localization is required.

For the magnetic fingerprints, there are three types of raw sensor readings available: $[B_x, B_y,$ and $B_z]$. In addition, we can easily calculate the norm of the magnetic vector $||B||$, which is the total strength of magnetic field. Fig. 2 shows an example of indoor magnetic fields constructed by the magnetic SLAM. It can be observed that the magnetic fingerprints have some power to discriminate the locations. However, using magnetic fingerprints alone is not sufficient since the magnetic fields are not bijective and have location ambiguities between similar fields. Moreover, because the true magnetic north is unknown, the available independent measurements fall into two elements: the total strength and vertical component (or total strength and inclination). To supplement the deficiencies of the magnetic fingerprints, we thus added another magnetometer in our robot system. The two magnetometers were installed at a specific distance from each other [Fig. 5(a)] and this increased the number of independent measurements. The insertion of an additional magnetometer also enhances the SLAM performance since it reduces the divergence caused by cell detection failures, especially when the robot moves along the cell boundaries—if one magnetometer fails, the other one can compensate for this. To further enhance the location uniqueness of the fingerprints, we also employed additional radio sources. In this paper, we used CSS-based ranging beacons for the additional radio fingerprints. Note that other types of radio sources, such as Wi-Fi, Bluetooth, and active radio-frequency identification (RFID) also can be used. Since we do not perform trilateration, it is not necessary to know the transceiver location. The transceivers are deployed just randomly in our application.

In the following sections, we describe the proposed relocation methods based on the MCL and MLP.

B. MCL-Based Relocation

MCL is one of the particle filter-based approaches applied to the localization problem. Since the RBPF also employs particle filter for the robot state update, the prediction and update equations of MCL are almost same as the ones used in the RBPF. The difference lies in that the MCL should rely on static map information while the RBPF can recursively update the map state.

In classic MCL algorithms, a constant error covariance is used for measurement update, assuming that the deviation of sensor measurement is independent of the measured location. However, this assumption is not valid for the noisy indoor measurements (especially for the radio measurements), and we should consider location dependency of the error covariance. For this, some statistics of the sensor measurements are also encoded in a grid level during the SLAM procedure. More specifically, the encoded statistics are the mean and standard deviation of the sensor readings from two magnetometers and from a number of radio-ranging beacons. As the robot stays in a kth cell, it gathers the sensor measurements and when it comes out of that cell the following fingerprint matrix $\boldsymbol{\Lambda}^k$ is constructed:

$$\boldsymbol{\Lambda}^k = \left[\boldsymbol{\lambda}_\mu^k \;\middle|\; \boldsymbol{\lambda}_\sigma^k\right] \tag{17}$$

$$= \left[\begin{array}{c|c} \boldsymbol{\mu}_m^k & \boldsymbol{\sigma}_m^k \\ \boldsymbol{\mu}_r^k & \boldsymbol{\sigma}_r^k \end{array}\right] \tag{18}$$

where m and r are the subscripts for the magnetic and radio data, respectively, and $\boldsymbol{\mu}_{(\cdot)}^k$ and $\boldsymbol{\sigma}_{(\cdot)}^k$ are the mean and

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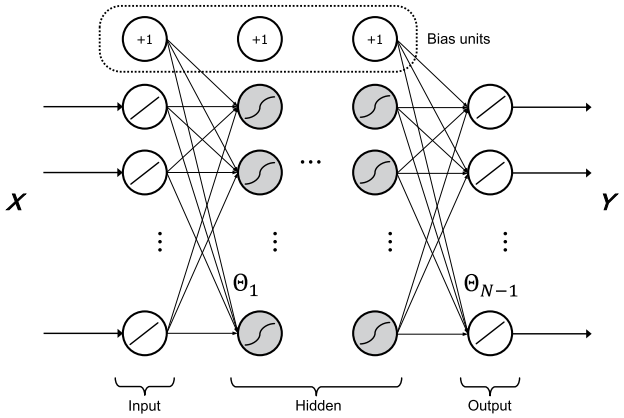


Fig. 3. Structure of the MLP with N layers. X and Y are the input feature vector and desired output vector, respectively. Θ_l is a matrix containing weight information of the lth layer's activation units.

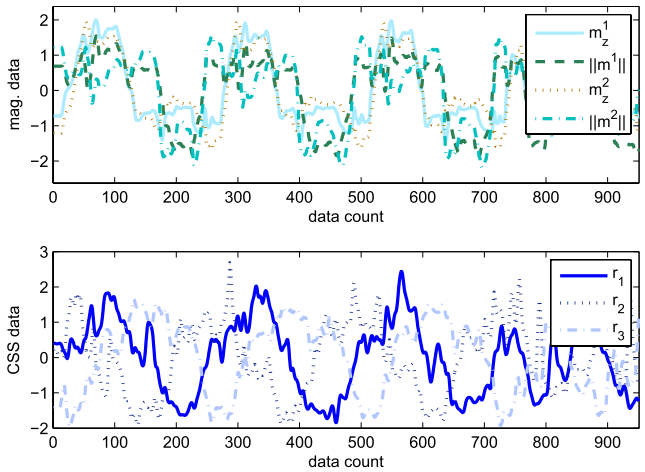


Fig. 4. Input features used in the MLP training. Four magnetic field observations (top) and three radio ranges (bottom) were used in the training. All the input features are normalized.

standard deviation of the measurement, respectively. The matrix elements are further described in the following:

$$\boldsymbol{\mu}_m^k = [\,\mu_{B_z^1} \quad \mu_{||B^1||} \quad \mu_{B_z^2} \quad \mu_{||B^2||}\,]^T \tag{19}$$

$$\boldsymbol{\mu}_r^k = [\,\mu_{r^1} \quad \dots \quad \mu_{r^q}\,]^T \tag{20}$$

$$\boldsymbol{\sigma}_m^k = [\,\sigma_{B_z^1} \quad \sigma_{||B^1||} \quad \sigma_{B_z^2} \quad \sigma_{||B^2||}\,]^T \tag{21}$$

$$\boldsymbol{\sigma}_r^k = [\,\sigma_{r^1} \quad \dots \quad \sigma_{r^q}\,]^T \tag{22}$$

where the superscript number in B is used to represent the two magnetometers and q is the number of radio-ranging beacons.

In the relocation phase, particles $\{\mathbf{x}_t^i\}$ with relevant weights $\{\gamma_t^i\}$ are initialized first. Each particle is then shifted by the robot kinematics and its weight is updated by the measurement $\boldsymbol{\zeta}_t$, which is defined as follows:

$$\boldsymbol{\zeta}_t = \begin{bmatrix} \mathbf{m}_L^1 \\ \mathbf{m}_L^2 \\ \mathbf{r} \end{bmatrix} \tag{23}$$

where $\mathbf{m}_L^j = [B_z^j, ||B^j||]^T$ is a vector related to the jth magnetometer readings and $\mathbf{r} = [r_1, \dots, r_q]^T$ is a vector of radio ranges. The weight update equation is then described as follows:

$$\gamma_t^i = \gamma_{t-1}^i \frac{1}{\sqrt{2\pi\left|\mathbf{R}_k^i\right|}} \exp\left[-\frac{1}{2}\left(\tilde{\boldsymbol{\nu}}_t^i\right)^T \left(\mathbf{R}_k^i\right)^{-1} \tilde{\boldsymbol{\nu}}_t^i\right] \tag{24}$$

where $\tilde{\boldsymbol{\nu}}_t^i = \boldsymbol{\zeta}_t^i - \boldsymbol{\lambda}_\mu^k$ is the measurement residual and $\mathbf{R}_k^i = \text{diag}([\boldsymbol{\lambda}_\sigma^k])$ is the noise covariance matrix, both with cell index k of the ith particle. The estimated robot position is then finally calculated as a weighted sum of the particles, similar to (16).

C. MLP-Based Relocation

A general structure of the MLP is shown in Fig. 3. The MLP structure has one input layer, more than one hidden layers, and one output layer. The hidden layer has sigmoid activation units and other layers have linear activation units. The numbers of input and output units are decided by dimensions of the input feature vector X and the desired output vector Y. In our case, X is the fingerprint measurements defined in (23) and Y is the 2-D robot position. Deciding the hidden structure of the model such as the number of hidden layers and the number of activation units in each layer is guided by the nonlinearity of the problem and fixed based on the validation data set.

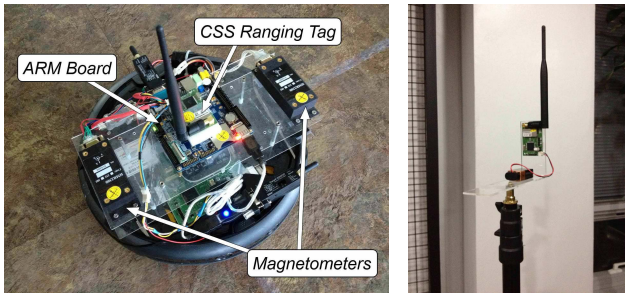
Training of the MLP is done by the conventional error backpropagation algorithm. We employed classic gradient descent and conjugate gradient methods as our optimization frameworks. Both methods require definition of the MLP cost function $J(\Theta)$, which is described as follows:

$$J(\Theta) = \frac{1}{2m} \sum_{i=1}^{m} \left|\left|Y^{(i)} - h_\Theta\left(X^{(i)}\right)\right|\right| \tag{25}$$

where Θ is a matrix containing weight information of MLP units, m is the number of training data sets, and $h_\Theta(\cdot)$ is the resulting MLP output. The training is done either in batch or stochastic mode. In batch mode, all the training data sets are used for a single update of the MLP parameters. In stochastic mode, a single set of the training data from a randomly chosen sequence is used for every single update. The sequence is regenerated by random permutation at every new epoch. Usually, the stochastic mode is faster than batch mode but it also has larger oscillations around the convergence region. This can be resolved by assigning a small value to the learning rate α. Further improvements can also be achieved by applying regularization, where the magnitudes of the MLP parameters are weighted in the cost function to prevent overfitting.

Fig. 4 shows the normalized input features used in the training procedure. The data have been obtained from the measurements performed to generate the simulation data, as explained in Section V-A. For the faster convergence, all the input features are normalized to have zero mean and unit variance. We have four magnetic field observations and up to three radio ranges as our input features. Note that the insertion of the another magnetometer increases independent measurements. The desired output was obtained from the resulting position from the magnetic SLAM.

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(a) (b)

Fig. 5. (a) Robot and (b) radio beacon. The robot is equipped with two magnetometers and one active radio ranging tag.

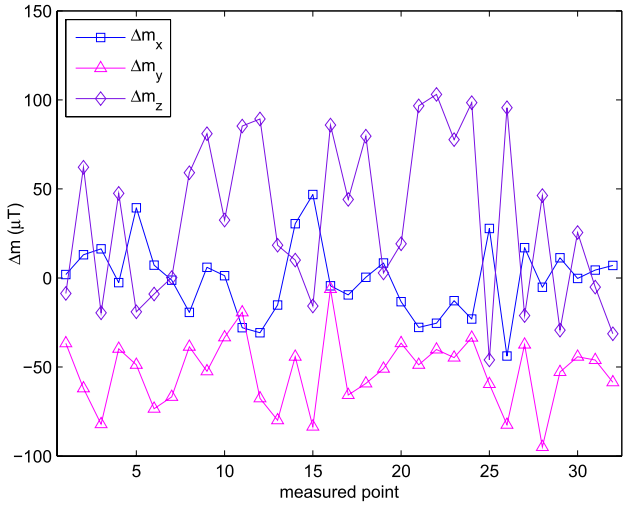


Fig. 6. Two magnetometer's bias difference. Since the robot hardware affects each magnetometer by different amounts of magnetic interference, we should calibrate it to use the magnetometer readings in our algorithms. The average and the variation of these bias differences were used in the magnetic field measurement model.

V. Validation of Proposed Algorithms

In this section, we evaluate the performance of the proposed algorithms by simulation analysis and real-world experiments.

A. Sensor Setup

Fig. 5 shows the mobile robot (Samsung, Tango) used in our study. The differential drive robot is equipped with two magnetometers (Honeywell, HMR 2300) and one active radio range sensor (Nanotron, nanoLOC TRX). Three radio beacons are installed around the area where the robot moves.

Since the robot hardware—especially, the motor part imposes different amounts of magnetic interference on each magnetometer; we first checked the bias differences between the two magnetometers. Fig. 6 shows the magnetic bias difference measured at 32 different test locations. The average of these bias differences was used in the compensation of the magnetic measurements. The variation of the bias measurements was also used as an additional uncertainty in the magnetic field measurements.

For the simulation data, the robot followed a window-shaped trajectory with two large outer loops and two smaller inner loops in an office building lobby [the same environment as in Fig. 13(a), but with a different path]. The average speed of the robot was 30 cm/s.

B. Simulation Setup

1) Emulating the Relocation: To emulate the situation of the relocation, we simply reuse the same data that are used for the SLAM. For MCL-based relocation, we run all the sequence of the data and save the converged location after every certain iteration. The particles are initialized whenever the convergence occurs. For MLP-based relocation, we separate the training data from the full data sequences. The relocation position is then calculated from each input set of the training data. For both MCL- and MLP-based methods, the relocation error is calculated by the difference between the SLAM and the relocated positions.

2) Parameter Setup: Here, we set several parameters applied in the simulation analysis. For the magnetic SLAM, the noise covariance values for the encoder and magnetometer readings were set to 0.015^2 $(\text{m/s})^2$ and 1.5^2 $(\mu\text{T})^2$, respectively. The particle size was set to 50 and the resampling threshold was set to 0.7 of the particle size. The cell width was set to 30 cm. For the MCL-based relocation, the covariance for the encoder reading was set as 0.015^2 $(\text{m/s})^2$. For comparison purpose, the particle size was varied from 10 to 500 and the cell size for the fingerprint construction from 10 to 50 cm. For the MLP-based relocation, the MLP structure of two hidden layers with 20 hidden units for each layer was used. The learning rate α was set to 0.5 for batch mode and 0.05 for stochastic mode. All these values were chosen by a separate validation test. The maximum epoch was set to 5000 for batch mode and 50 for stochastic mode. No regularization was used in our training procedure since it did not have a critical effect with our data set. For both MCL- and MLP-based relocation, we also varied the number of available sensors to verify the enhancement by additional radio sources.

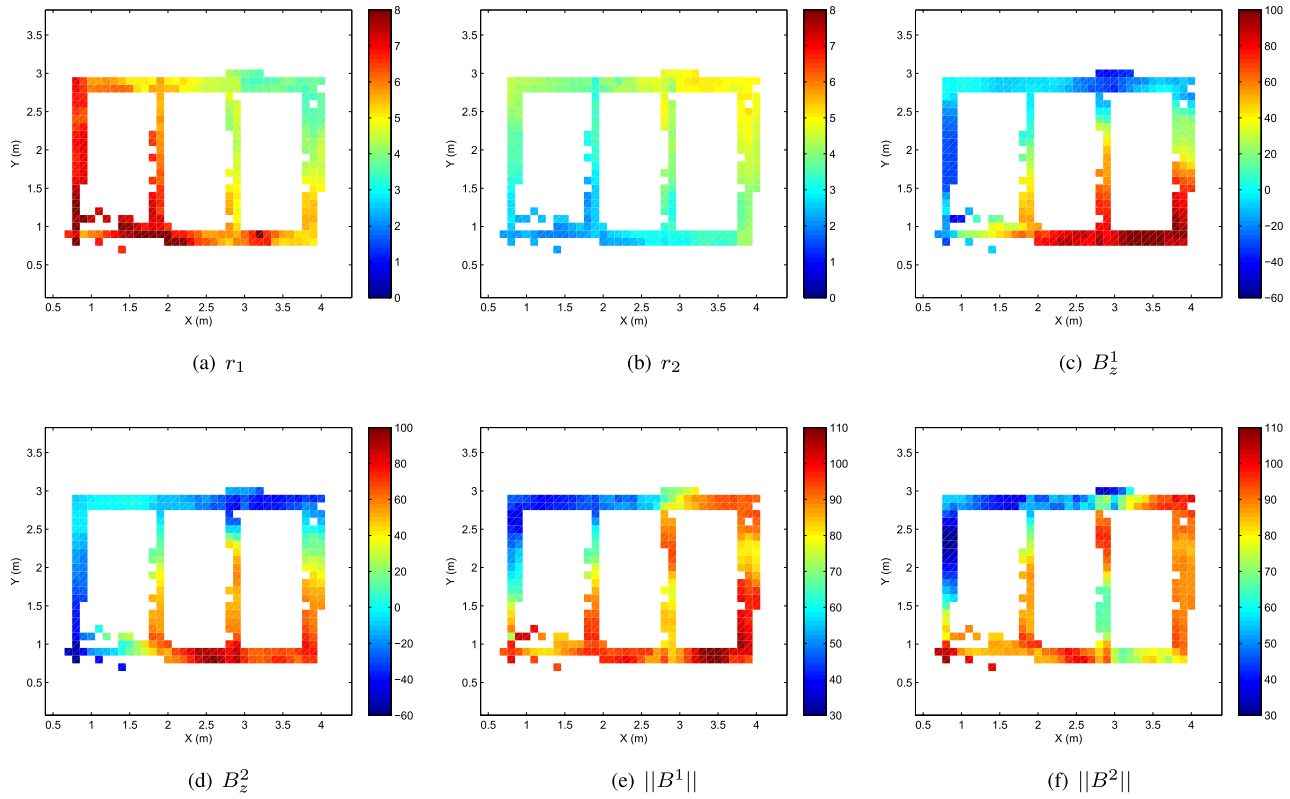
C. Simulation Results

1) Magnetic SLAM: Figs. 7 and 8 show the constructed fingerprints encoded with the estimated SLAM locations. Note that the robot's heading is finely corrected without any gyroscopic devices. This verifies that even if the global geomagnetic field suffers from anomalies, local corrections in orientation using the magnetic landmarks can be achieved. The SLAM results are further described later in the experimental results.

2) MCL-Based Relocation: The constructed fingerprints $\boldsymbol{\lambda}_{\mu}^{k}$ in Fig. 7 show that both the magnetic and CSS radio fingerprints are highly nonlinear in the spatial domain, even though the test environment in Fig. 13(a) has not much of the obstacles. In Fig. 8, we can also observe that the variance of the fingerprints $\boldsymbol{\lambda}_{\sigma}^{k}$ varies depending on the locations, which is reflected in our MCL weight update equation (24).

Fig. 9 shows the MCL results with different parameters. The MCL performance is clearly better with a larger particle size. In Fig. 9(b), we can observe that the performance is rapidly degraded with a cell size of less than 10 cm.

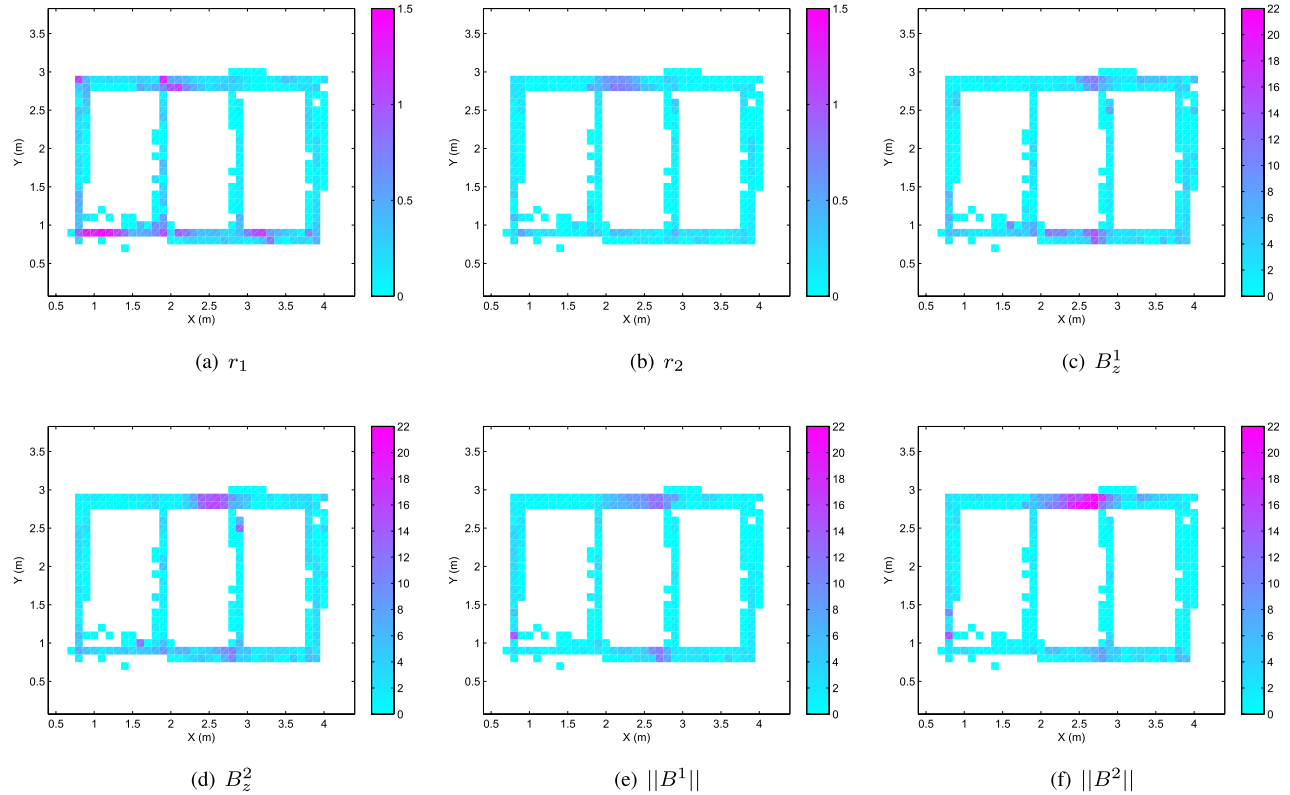
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(a) r_1 (b) r_2 (c) B_z^1

(d) B_z^2 (e) $||B^1||$ (f) $||B^2||$

Fig. 7. Constructed fingerprints $\boldsymbol{\lambda}_{\mu}^{k}$ (mean) encoded with the estimated SLAM locations. (a) and (b) CSS fingerprints. Units are in meters. (c)–(f) Magnetic fingerprints. Units are in μT.



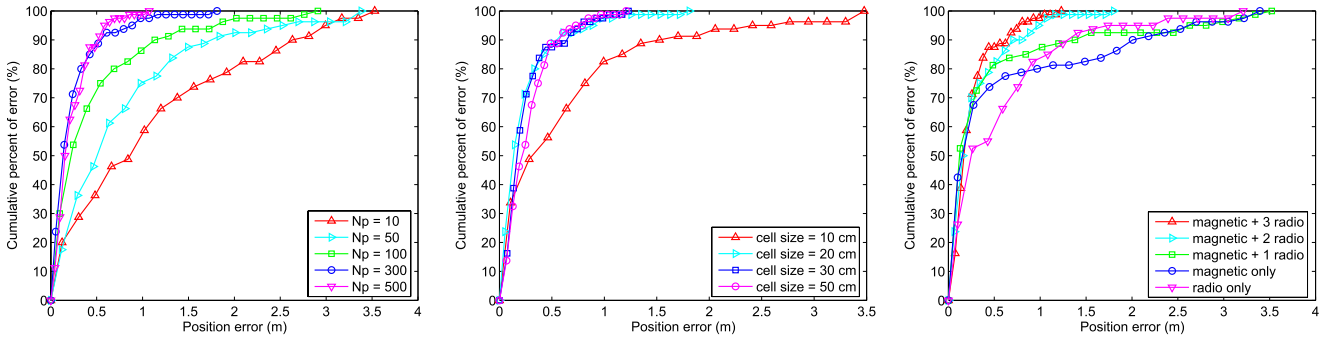
(a) r_1 (b) r_2 (c) B_z^1

(d) B_z^2 (e) $||B^1||$ (f) $||B^2||$

Fig. 8. Constructed fingerprints $\boldsymbol{\lambda}_{\sigma}^{k}$ (standard deviation) encoded with the estimated SLAM locations. (a) and (b) CSS fingerprints. Units are in meters. (c)–(f) Magnetic fingerprints. Units are in μT.

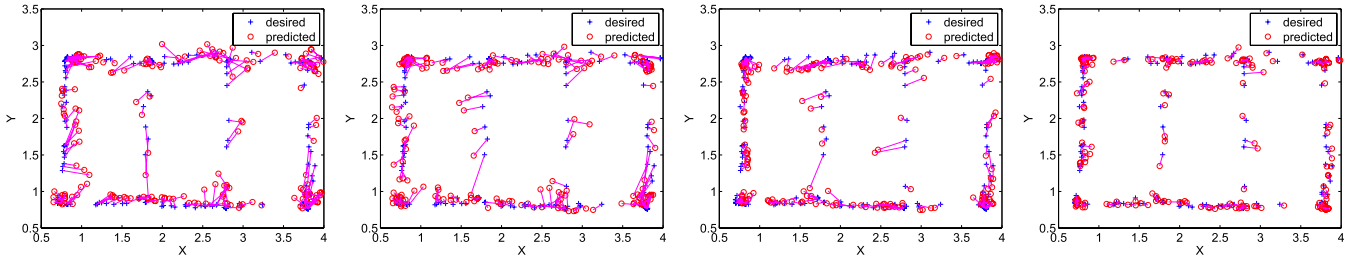
This can be explained by the fact that in smaller cells there is less likelihood of gathering enough samples to describe the statistical fingerprint values. Fig. 9(c) shows the performance difference in the MCL results with a different sensor set. As we pointed out previously, the location fingerprints can be improved by combining magnetic and radio sources.

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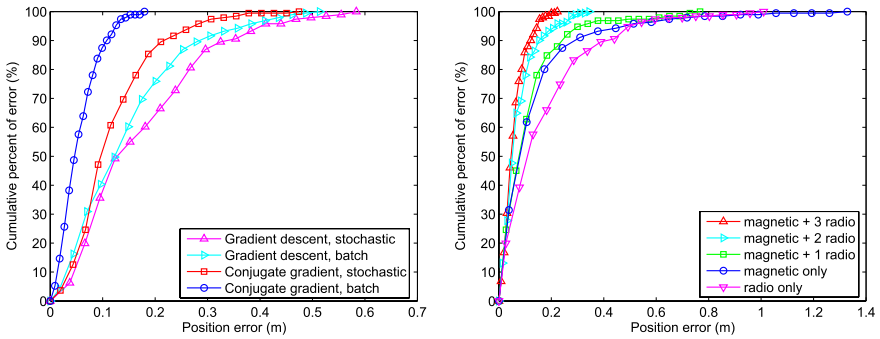
(a) (b) (c)

Fig. 9. MCL relocation results with varying parameters. (a) Results with different particle sizes. The cell size was set to 20 cm and the number of radio ranging sensors was set to 3. (b) Results with different cell sizes. The particle size was set to 300 and the number of radio ranging sensors was set to 3. (c) Results with different sets of available sensor readings. The particle size was set to 300 and the cell size was set to 30 cm.



(a) (b) (c) (d)

Fig. 10. MLP-based relocation results from various training schemes. (a) Gradient descent, stochastic. (b) Gradient descent, batch. (c) Conjugate gradient, stochastic. (d) Conjugate gradient, batch.



(a) (b)

Fig. 11. MLP relocation errors for (a) various training schemes and (b) different number of available sensors.

More than 95% of the relocated position reached submeter accuracy after more than two radio beacons were applied.

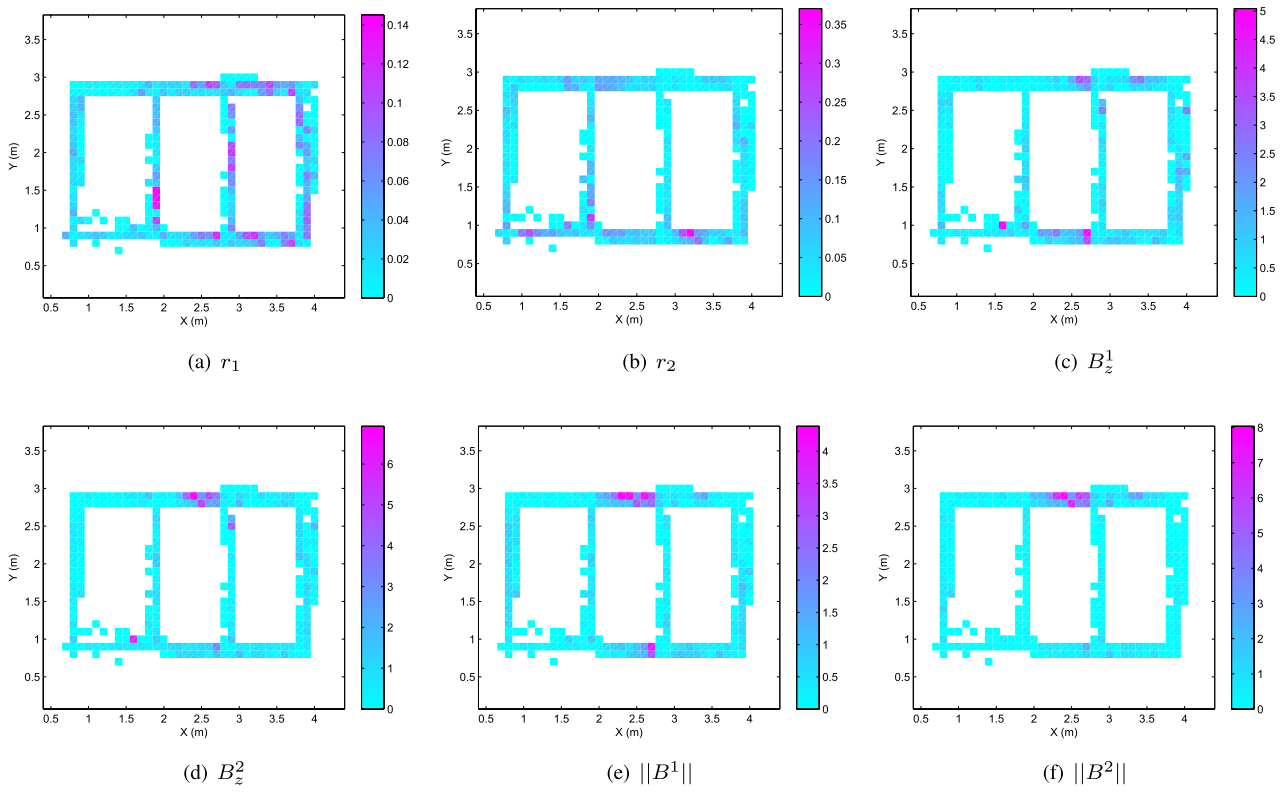
3) MLP-Based Relocation: Figs. 10 and 11(a) show the MLP relocation results from various training schemes. We can observe that conjugate gradient outperforms the classic gradient descent both in batch and stochastic modes. For the batch and stochastic modes, the two approaches show some tradeoff between the training time and performance. The batch mode usually outperforms the stochastic, but the stochastic mode can obtain a certain level of accuracy in a shorter time—in our case, training epoch was 1% to that of the batch mode. Fig. 11(b) shows the performance difference in the MLP results with application of a different sensor set. Here, we can also see that the combined feature set improves the relocation performance.

4) Uncertainty Analysis: Based on the constructed fingerprints, we can analyze the cell-level uncertainty of the magnetic and radio measurements. The standard Type A uncertainty measure U [36] is used in our analysis, which is calculated as follows:

$$U = \frac{s}{\sqrt{n}} \tag{26}$$

where s is the standard deviation, and n is the number of measurements. The constructed uncertainty values are shown in Fig. 12, where the number of measurements used was up to 134 for each cell. We can observe that the uncertainty patterns of magnetic measurements are similar to those of the standard deviation shown in Fig. 8, both showing large variations in the upper and bottom middle areas. In contrast, the uncertainty of the radio measurements is rather evenly distributed and small.

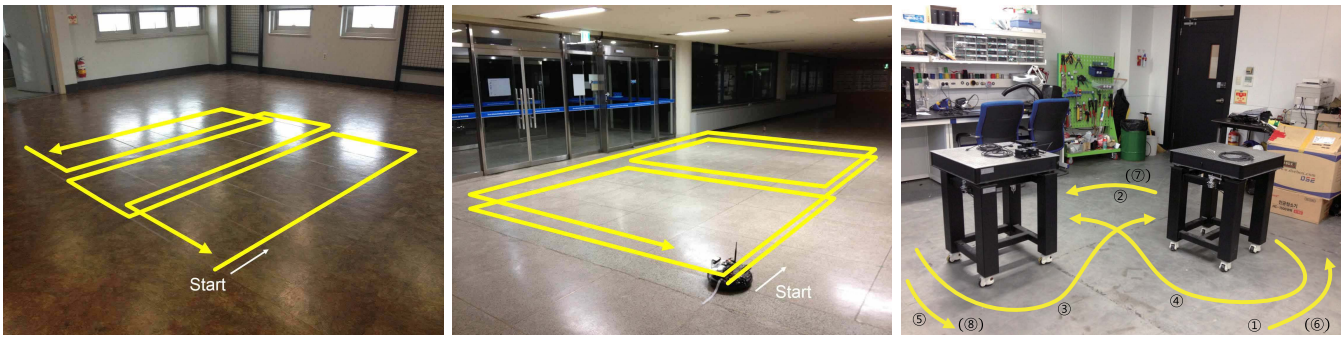
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(a) r_1 (b) r_2 (c) B_z^1

(d) B_z^2 (e) $||B^1||$ (f) $||B^2||$

Fig. 12. Uncertainty of the magnetic and radio measurements encoded with the estimated SLAM locations. (a) and (b) Radio measurements. Units are in meters. (c)–(f) Magnetic measurements. Units are in μT.



(a) (b) (c)

Fig. 13. Test environments with drawings describing the robot paths. (a) Robot starts and takes a S-shaped path until it returns to the starting point. It repeats this path twice and then the relocation starts. (b) Robot follows the rectangular-shaped loops drawn. The relocation starts after finishing this path. (c) Robot rather freely moves around in a lab environment filled with many steel objects. Then, the relocation starts.

TABLE I
COMPARISON OF MCL- AND MLP-BASED RELOCATION PERFORMANCE WITH A DIFFERENT SET OF AVAILABLE SENSOR READINGS

Available Sensor Readings	MCL		MLP	
	Mean (m)	STD (m)	Mean (m)	STD (m)
Radio only	0.619	0.657	0.190	0.177
Magnetic only	0.641	0.893	0.160	0.181
Magnetic + 1 radio	0.528	0.805	0.126	0.128
Magnetic + 2 radio	0.315	0.328	0.080	0.065
Magnetic + 3 radio	0.272	0.237	0.062	0.043

TABLE II
ATE METRIC COMPARED FOR THE DIFFERENT ESTIMATION METHODS. ALL UNITS ARE IN METERS

Data Set	Odometry	SLAM (EKF)	SLAM (RBPF)
Fig. 13(a)	0.304 $\pm$ 0.234	0.093 $\pm$ 0.056	0.041 $\pm$ 0.025
Fig. 13(b)	0.574 $\pm$ 0.482	0.185 $\pm$ 0.139	0.059 $\pm$ 0.036
Fig. 13(c)	0.408 $\pm$ 0.152	0.272 $\pm$ 0.095	0.226 $\pm$ 0.033

D. Implementation on an Embedded System

Table I summarizes the performance of the MCL- and MLP-based relocation algorithms with a different set of available sensor readings. In all cases, the MLP-based relocation outperforms the MCL-based relocation. Thus, we decided to implement the MLP-based relocation along with the SLAM algorithm on an embedded system (ARM11). The implemented system was then tested in three different environments,

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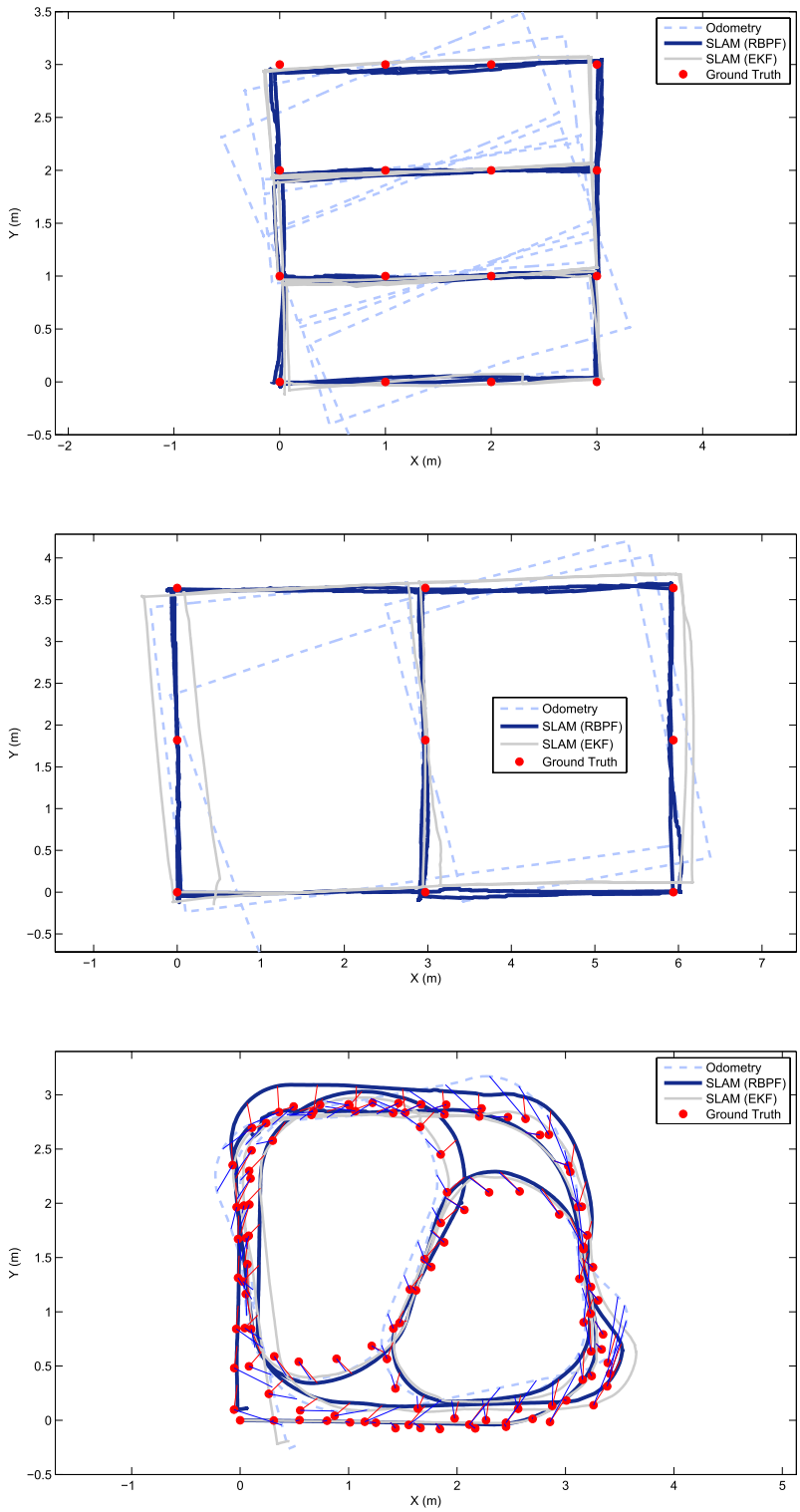


Fig. 14. Comparison of raw odometry and the proposed magnetic SLAM results. EKF-SLAM results with same prediction and measurement models [(5) and (6)] are also provided. Each of the results are from the tests performed in the environments shown in Fig. 13. The SLAM trajectory obtains correct orientations while the odometry fails due to the wheel encoder noise. In terms of the SLAM framework, the RBPF shows consistently better performance than EKF.

as shown in Fig. 13. In this online test, the robot was first driven along the given paths performing magnetic SLAM. After finishing the given trajectories, the robot was lifted and resituated at the predefined relocation points. The robot lift was detected by the lift sensor.

The resulting SLAM trajectory and the relocation results are shown in Figs. 14 and 15, respectively. In the SLAM results, we also provided trajectories using a standard extended Kalman filter (EKF) SLAM with same prediction and measurement models [as defined in (5) and (6)]. We can see that

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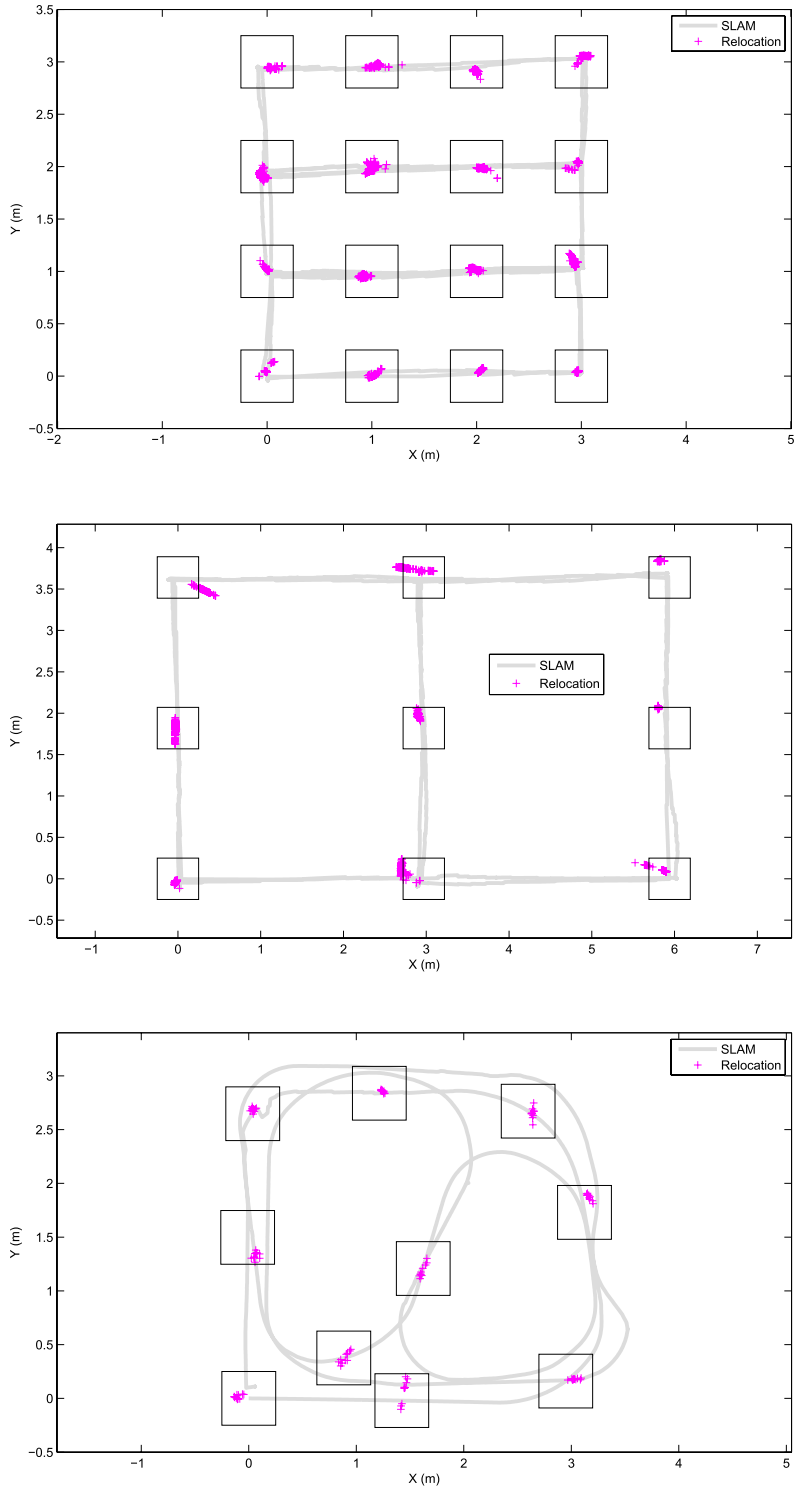


Fig. 15. MLP relocation results. Each of the results are from the tests performed in the environments shown in Fig. 13. Gray line represents the SLAM path where the robot gathered MLP training data for the relocation and cross-shaped marker represents the prediction of the relocated robot position. For quick evaluation, squares of 0.5-m length were centered at the ground positions where the robot was actually repositioned. The rmses for the three tests were 0.06 ± 0.03, 0.18 ± 0.11, and 0.07 ± 0.01 m, respectively.

the magnetic SLAM obtains correct orientations without any help of heading devices. This is because the robot heading state θ_t is locally corrected by the measurement Jacobian $\mathbf{H}_m$, which affects the calculation of the innovation covariance matrix $\mathbf{S}_t$ and thus the weight update equation (15) (or the Kalman gain matrix in case of EKF-SLAM). Table II summarizes the absolute trajectory error (ATE) for the resulting trajectories determined by different methods. In our test data, the RBPF shows consistently better performance than EKF. In the relocation results, we can see that the estimated relocation

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positions correctly catch the past SLAM trajectories. The root mean square error (rmse) for the estimated position for each data set was 0.06 ± 0.03, 0.18 ± 0.11, and 0.07 ± 0.01 m, respectively.

VI. Conclusion

In this paper, we proposed a method of solving indoor SLAM and relocation problems by exploiting ambient magnetic and radio sources. The proposed magnetic SLAM was realized in the RBPF- and grid-based SLAM frameworks, where we exploited the local heading corrections by the magnetic measurements. For the relocation, MCL- and MLP-based methods were explored and the simulation analysis with real-world data compared the performance of the two algorithms, where the MLP-based approach outperformed the MCL-based method. We also implemented the proposed magnetic SLAM and relocation algorithms in an embedded system and verified the feasibility of the proposed methods as an online navigation system for mobile robots.

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